

How much does the map of science depend on the embedding model?

Alejandro Treny Ortega

Independent researcher

Voice option 3: Distinctive and reflective • 19 September 2026

Abstract

A field comparison can survive repeated article selection and still reverse when the embedding model changes. This study examines that problem using ten models and a fixed corpus of 500,000 OpenAlex records from 26 fields and 2000–2024. Matched panels compare field centres, internal relationships and nearest neighbours, with checks on text input and representation settings. The models share broad structure, yet retain only 30.3% of the 25 nearest neighbours on average within field-period groups. Removing abstracts changes neighbours about as much as changing models; removing titles changes less. A later exploratory analysis examines all 325 field pairs. Persistent opposite orderings across models occur in 262 pairs for angular spread and 221 for effective linear dimension; requiring relative differences above 5% leaves 96 and 177. Alternative measures preserve many oppositions, although centring and pooling affect the conclusions. The results describe geometric dependence, without establishing thematic correctness. I argue for following a claim through the choices that produce its map. Repeating a result under article selection answers a useful question, but it does not establish that another representation of the same papers supports the same interpretation.

Keywords: science mapping; document embeddings; robustness; neighbourhood agreement; geometric comparisons.

1 Introduction

Changing a document representation can reverse a comparison between research fields while the articles stay the same. That is the difficulty behind this study. A science map is useful because it turns a collection of papers into relationships that can be inspected: nearby work, neighbouring areas and contrasts between disciplines. With text embeddings, a model first translates each paper into a vector. The distances and arrangements available for interpretation come from those vectors. Applications to biomedical literature and physics show what this makes possible (González-Márquez et al., 2024; Constantino et al., 2025). I ask how far a conclusion drawn from such a map travels when the embedding model changes.

The difficulty is older than these encoders. Gläser et al. (2017) asked what a reconstructed thematic structure owes to scientific knowledge and what it owes to the method of reconstruction. Boyack and Klavans (2020) found that text, citation and hybrid maps could have similar aggregate accuracy while assigning substantially different papers to their clusters. Those results locate the problem in the interpretation of a comparison. A favourable global score does not tell a reader whether the particular grouping they want to use survives a change of method.

I examine this with 500,000 publications labelled English in OpenAlex, published in 2000–2024 and assigned to 26 broad Fields. The papers remain fixed across ten embedding models. Field centres, relationships within fields and individual article neighbours provide successive objects of comparison. Matched subsets then change the text supplied and the rule that turns model output into one vector per paper. Later, an exploratory geometric extension follows a more pointed question: whether two fields retain their ordering when the representation changes, after checking what happens when the articles change.

The models do share broad relationships. The local comparisons and geometric field orderings show how much work remains before using that agreement to support a particular claim. One model can repeatedly give one ordering as articles change, while another repeatedly gives the reverse. I treat the size of that difference and its response to processing as part of the result. A direction can be persistent and its magnitude small. The comparisons reveal which relationships depend on the choices tested; deciding whether a relationship is substantively right requires another kind of validation.

2 Background and research questions

2.1 Representation choices in science mapping

Earlier evaluations are useful precisely because they state what a representation is being asked to do. Constantino et al. (2025) compared text and graph representations with a hierarchical classification of physics papers, including inputs based on titles and abstracts. In a biomedical literature map, González-Márquez et al. (2024) compared models and output-reduction rules through the field labels shared by nearby papers. These are concrete evaluation criteria. They clarify why agreement across models, which is the concern here, cannot silently take the place of an external assessment.

There is also more than an encoder inside a document representation. Someone has to decide which text it receives, how much of that text fits and, for some models, how its token outputs become a single vector. Pooling is this last decision: taking an average and selecting a designated output position need not preserve the same relationships. The sensitivity to short queries and wording changes documented by Singh and Singh (2022) makes the input

issue tangible. I include these choices because a model comparison otherwise leaves part of its own procedure unexamined.

The joint comparison of geometry and neighbours is already part of the literature. Caspari et al. (2024) used it across retrieval datasets, and Imel and Hafen (2025) examined density and asymmetry across document embeddings and disciplines in relation to citations. The question I pursue is what happens to broad, local and geometric conclusions when those comparisons are made within a fixed panel. Matched articles, explicit text and representation conditions, and controlled neighbour candidates keep each result attached to the change actually tested.

2.2 Agreement, stability and interpretation

I avoid treating robustness as a property that can be assigned to the whole map in one step. Agreement asks what two representations share on the same papers. Stability asks whether a given result survives a given change. A stable ordering within each model can still be an ordering on which the models disagree. Thematic validity goes further and asks how those relationships correspond to the research itself. In invasion biology, Held and Velden (2022) showed why interpreting algorithmic groups requires attention to the relationships a mapping method combines. A repeated answer or agreement between models cannot do that interpretive work alone.

The word *shape* needs similar care. Here it refers to relationships in the embedding spaces before projection into two dimensions. Centres summarise the arrangement of fields; articles retain their internal organisation; neighbours identify the papers closest to a particular article. Angular spread describes the separation of vectors, and effective linear dimension describes how variation occupies directions. Each is a limited view of the representation. Calling a field more spread out does not establish that its topics are more diverse; assigning it more effective dimensions does not count its topics. Keeping the object explicit makes a change in shape interpretable.

2.3 Research questions and scope

Three questions organise the analysis:

- RQ1.** Which structural relationships are shared across embedding models, and how does agreement vary between field centres, articles within fields and local neighbours?
- RQ2.** How much do text input and pooling choices change geometry and neighbours relative to changing the embedding model?
- RQ3.** Which geometric comparisons between fields persist, reverse or remain unresolved across models and article selections, and how large are the differences?

These questions concern the corpus and models actually compared. The third was added after the initial results, in the exploratory extension documented in the methods and supplement. I retain that order of development. OpenAlex labels organise the comparisons and remain in place with their known errors; they are not an independent thematic reference. The study can establish persistence within this design. It cannot identify a single correct map and then use that map to declare a winning model.

3 Data and comparative design

3.1 Corpus and analytical panels

The corpus contains 500,000 distinct OpenAlex work IDs, collected and cleaned by 15 September 2026. OpenAlex provides the metadata and texts (Priem et al., 2022); its core catalogue was restricted to articles, reviews and conference papers published in 2000–2024, labelled English, with a non-empty title, a reconstructible abstract of at least 50 words and a known primary Field. Retraction and paratext flags were excluded. Clear language and content failures were removed, while uncertain cases and possible duplicate versions were retained with flags. Each record’s primary-topic classification was frozen at extraction. These automatically assigned labels use text, citation and journal information (OpenAlex, 2026); they organise the analysis rather than supply an independent standard of correctness.

A general sample of 400,000 eligible records was drawn through seeded random API requests. An additional 100,000 records reinforced the smallest Field-period groups by repeatedly allocating places to the least populated group. Large groups were not reduced to the size of the smallest. The resulting 130 groups cover 26 Fields and five periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Every group contains at least 2,165 records. Global corpus comparisons use the general sample; group comparisons can use both cohorts. Equal-weight averages across groups describe a balanced comparison, not the proportions of world research.

Using the same papers is a starting condition; the different questions also need different panels (Table 1). Three text inputs are compared on 52,000 articles balanced by Field and period, including the earlier 26,000-article pilot. A separate 52,000-article panel tests native inputs against a common literal fragment that fits every tokenizer. Equal-size group comparisons and matched neighbour searches control further changes in the comparison. Every model comparison retains the same article identities. Supplement S1 gives the extraction and cleaning rules, the counts and the flags that remain.

Table 1: Corpus and analytical panels.

Panel	Articles and representation	Comparison and scope
Original corpus	500,000 IDs: 400,000 base + 100,000 supplement. Ten models; title + abstract.	26 Fields, five periods. Global summaries use the base; the full corpus is not population-proportional.
Three text inputs	52,000 IDs; 400 per Field-period, 2,000 per Field. Title, abstract and both.	Same articles and candidates between models. Includes the 26,000-article pilot.
Common fragment	A different 52,000-ID panel; 400 per Field-period.	Paired native/shared content; model-specific tokenization. Distinct from the three-input panel.
Matched group size	256 articles per centre; 26 Fields, 217 Subfields, or 26 Subfield centres.	Within groups: 26 Fields and the same 183 Subfields at 128, 256 and 512 articles. Different objects from centres.
Paired searches	217 Subfields; 50 fixed queries each (10,850 total). Ten candidate selections.	256 candidates in each scope, including all 50 queries. The Field pool is conditioned on these queries.
Geometry / Field pairs	2,000 per Field; twenty halves of 1,000 and five selections of 2,000.	325 Field pairs and two properties; 26 saved conditions. Alternative metrics have different coverage.

Rows overlap and must not be added. Periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Each model comparison uses the same IDs. Query articles exclude themselves during neighbour search. The 1,300 technical test articles are not an additional scientific panel. Full filters and quotas: Table S1.

3.2 Models and text representations

I chose a fixed panel of ten models to compare representations used for scholarly material across different training approaches (Table 2). It includes scientific-document, general sentence and word-level biomedical/scientific models. Its purpose is a bounded comparison; it cannot represent a random population of encoders or rank the latest releases. Checkpoints, tokenizer revisions and text formats were frozen before inference. Models represented the same titles and abstracts with their customary format and native limit. MiniLM’s 256-token setting is primary, and a separate 512-token variant checks its sensitivity. Native limits truncated 49.58% of MiniLM inputs and 20.16% of MPNet inputs.

For BERT, SciBERT, BioBERT and PubMedBERT, the principal document vector is the mean of last-layer, non-padding token states, including special tokens. These models were not trained specifically for document similarity. Their first-position state (CLS, before the pooler) and final non-padding state (SEP) provide alternative readouts. The mean rule was specified before the primary comparisons and retained after the early sensitivities; it was not selected to maximise agreement. The other six models use their documented representation recipes. Title-only and abstract-only inputs use the same 52,000 IDs as the combined-input comparison. Native limits mean that this tests practical ways of using the models, without isolating architecture from text length.

Table 2: Embedding models and primary representation settings.

Model	Dim.	Tokens	Readout	Reference
SPECTER	768	512	CLS	(Cohan et al., 2020)
SPECTER2	768	512	CLS + adapter	(Singh et al., 2023)
SciNCL	768	512	CLS	(Ostendorff et al., 2022)
SciBERT	768	512	Mean	(Beltagy et al., 2019)
BERT	768	512	Mean	(Devlin et al., 2019)
MPNet	768	384	Mean	(Reimers and Gurevych, 2019)
MiniLM	384	256	Mean	(Reimers and Gurevych, 2019)
PubMedBERT / BiomedBERT	768	512	Mean	(Gu et al., 2022)
BioBERT	768	512	Mean	(Lee et al., 2020)
SimCSE	768	512	CLS, pre-pooler	(Gao et al., 2021)

MPNet and MiniLM use all-mpnet-base-v2 and all-MiniLM-L6-v2, respectively. Limits count tokens including special symbols. Mean: last-layer average over non-padding positions, including special symbols. CLS: first-position state before the pooler; SPECTER2 adds its proximity adapter. For MPNet/MiniLM, the reference describes the Sentence-BERT framework, not the exact check-point training. Readout variants are not additional models. Full revisions, formats and truncation: Table S2.

3.3 Objects and measures

All measurements use the original embedding dimensions, without a two-dimensional projection. Article vectors are first scaled to unit length. Shape agreement is measured primarily by corrected linear centred kernel alignment (CKA), which compares relationships among the same identified articles (Kornblith et al., 2019; Murphy et al., 2024). Higher values indicate greater agreement. The finite-sample correction reduces a known source of inflated alignment; it does not make the complete ratio an exactly unbiased estimator. Procrustes

alignment and correlations between distance ranks provide checks with different sensitivities. Their definitions and the correction formula appear in Supplement S2. These related measures are not independent votes about validity (Williams et al., 2021; Williams, 2024).

A centre compresses the unit-length article vectors of a group into their mean. That is the object used to compare broad organisation. The within-group analysis keeps the articles separate. I use random groups to examine how much centre agreement can accompany the act of averaging without the observed labels. Group sizes and publication periods are preserved, and every model receives the same random allocation.

Neighbour agreement is the fraction of the k closest articles shared by two models. The main choice is $k = 25$, with 10 and 50 as checks. Searches use cosine similarity and exclude the query article itself. The candidate pool is stated for each result: all articles in a Field-period group, an equal-size subset, a matched Field/Subfield pool, or the general corpus. An exact blockwise calculation avoids storing every pairwise distance. Independent full-sort checks verify saved neighbour lists. Neither overlap nor its simple chance-adjusted form removes all effects of candidate composition or size.

Two additional properties describe each Field geometrically. *Angular spread* is the median angle between distinct article vectors. *Effective linear dimension* is the participation ratio of the centred covariance matrix C , $PR = (\text{tr } C)^2 / \text{tr}(C^2)$. It is larger when variation is distributed across more directions. Angular quantiles, angles to the centre, entropy-based effective rank and the number of directions containing 80% of variance serve as alternatives. The latter two use the same covariance spectrum as PR; they do not provide independent thematic validation. Graph-connection measures were also tested, but their synthetic counterexamples prevented an interpretation as general fragmentation (Supplement S6).

3.4 Matched comparisons and stability

The strict Field/Subfield neighbour comparison fixes 50 query articles in each of 217 Subfields. Both searches contain 256 candidates, including those same 50 queries, and identical period counts. Ten selections vary the remaining 206 candidates within the Subfield or its parent Field. Thus, the broader search is conditioned on the queries; it is not an unrestricted draw of 256 Field articles. The design covers 10,850 queries and holds search opportunity constant while varying its scope.

For Field geometry, I follow all unordered pairs of the 26 Fields: 325 comparisons for each property. A model supports a persistent direction only if that direction survives the primary 2,000 articles per Field, twenty half samples of 1,000 and five additional selections of 2,000 from the same corpus. All-ten agreement requires the same persistent direction throughout the panel. Opposition requires at least two fixed models that repeatedly support opposite directions. The remaining pairs are unresolved, which leaves equality unestablished. To distinguish a change of direction from the size of a difference, relative magnitude is $2(B -$

$A/(A+B)$, where A and B are the Field values. Minima of 0, 1, 5 and 10% test sensitivity to small differences.

These repetitions stay inside one frozen corpus. They test sensitivity to article selection; their overlapping subsets supply neither independent population samples nor confidence intervals. I retain excessive median shifts and wide selection ranges as alerts, with the operational rules and denominators recorded separately in the supplement. Pair counts need the same restraint. Models recur across model pairs, and Fields recur across Field pairs, so these counts are descriptive. No significance test treats the pairs as independent observations.

3.5 Sequence of checks

The sequence of the work is part of its interpretation. The initial shape and neighbour comparisons came before the larger input panel, matched Subfield searches and case atlas. Morphology was a later exploratory extension, and its all-pairs summary followed the first examples. Rules were recorded locally before the corresponding new calculations. This record is not an external preregistration of the study as a whole. Random-group centres and the fixed-neighbour-fraction check were among controls added after earlier results. Supplement S2 records the sequence, and Supplements S3–S7 retain the time, composition, model-trait, source-error and processing checks, including those that narrow what can be concluded.

4 Results

4.1 Shared broad structure and within-field variation

Models agree strongly about the arrangement of group centres. With 256 articles contributing to each centre, mean corrected CKA is 0.939 for the 26 Fields and 0.911 for 217 Subfields (Figure 1A). Restricting the latter to 26 Subfield centres gives 0.902. Random groups with matched sizes and dates also show positive agreement, around 0.63–0.64, but do not recover the observed level. Averaging therefore contributes to the comparison, yet is insufficient to account for the agreement associated with these labelled groups.

The internal organisation has a more modest level of agreement: mean CKA is around 0.63 for Fields and 0.64 for the same 183 Subfields eligible at every size (Figure 1B). Those averages move little between 128 and 512 articles. A finer disciplinary label thus does not automatically yield a less consistent comparison. There is a large gap from the centre panel, but its interpretation stops short of a causal effect of scale. Averaged centres and their individual articles are different objects.

The alternative measures broadly support the ordering of model-pair agreements. Across the original 130 Field-period groups, the mean rank correlation with corrected CKA is 0.891

for Procrustes and 0.881 for distance ranks. Their agreement is incomplete, and the original selection screen retains 50 alerts among 5,850 model-pair/group comparisons (Table S4). These qualifications concern particular comparisons; an overall mean does not erase them.

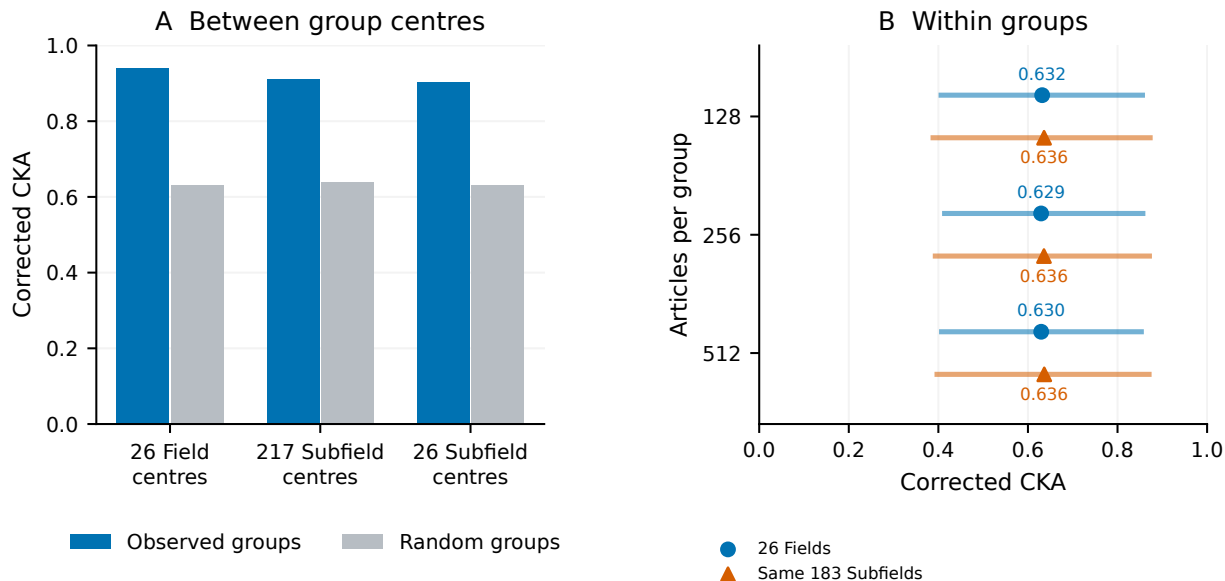


Figure 1: Agreement between group centres and within groups. (A) Mean corrected CKA across the fixed model pairs and saved centre comparisons; each centre contains 256 articles. Random groups preserve sizes and dates. The 26-Subfield condition controls the number of centres. (B) Within-group agreement at three article counts, for 26 Fields and the same 183 eligible Subfields. Points and labels show means; lines span the 2.5th–97.5th percentiles across group–model-pair comparisons (1,170 for Fields; 8,235 for Subfields). These are descriptive spreads, not confidence intervals; model pairs are not independent. All CKA axes span 0–1. The panels compare different objects, so their difference is not an estimated causal effect.

4.2 Neighbour agreement under matched comparisons

Across the full Field-period searches, two models retain an average of 30.3% of the 25 nearest neighbours: about 7.6 articles out of 25. This average gives each group and model pair equal weight. Restricting every group to 2,048 common candidates raises overlap to 31.9%. In the separate search across the 400,000-article general corpus, 13,000 balanced queries retain 17.5%. The last result changes both the scope and the number of available candidates, so it cannot by itself establish weaker agreement across disciplinary boundaries.

The matched Field/Subfield experiment addresses that ambiguity directly (Figure 2). With the same queries, dates and 256-candidate size, Subfield agreement is lower in 127 of 217 cases and higher in 90. The mean difference is small, at -1.35 percentage points. For 50

neighbours, the split becomes 109 lower and 108 higher. A narrower search can preserve more neighbours or fewer, depending on the group and the chosen neighbourhood size. There is no universal loss of agreement from Field to Subfield.

Shape and neighbour agreement do move together: the original 45 model-pair means have a descriptive rank correlation of 0.920. I keep both because the interpretation changes with the object. CKA summarises relationships spread through the space. A neighbour list names the papers a reader would meet there. Their association does not make those consequences interchangeable, nor does reporting both create independent evidence of reliability.

The atlas makes those local differences inspectable. Its Subfield examples are selected by a recorded persistence rule across 27 combinations of size, readout and neighbour count. Article-level examples then show links that persist across models alongside links whose inclusion varies sharply (Supplement S4). Some extreme cases expose source problems: a history book appears under Physics, an infection-related record under Political Science, and an announcement remains in the corpus. Comparison with the original records confirms that these are source or classification issues, rather than mismatched embedding rows. The examples are exploratory and deliberately selected; they estimate neither the prevalence of such errors nor the accuracy of the links.

4.3 Text input and representation choices

The text supplied to the model deserves as much attention as the model name. On the same 52,000 papers, changing encoders gives a mean shape change of 0.370; retaining the encoder and supplying only the title gives 0.366, with change measured as one minus corrected CKA. The corresponding fractions of neighbours replaced are 68.8% and 70.7% (Figure 3). An input choice can therefore have consequences of comparable size in this design. The proximity of the averages leaves equivalence untested.

Removing only the title changes less. Abstract-only inputs have a mean shape change of 0.027, yet replace 22.1% of the 25 neighbours. A small change in overall geometry can therefore still alter the papers returned locally. The model-specific results also vary substantially: title-only neighbour replacement ranges from 51.7% for MPNet to 84.2% for SciBERT (Table S5). A single pooled mean would conceal those differences.

Even the ordering of those average effects depends on the readout. With the principal mean rule, removing the title has a relatively modest effect and removing the abstract slightly exceeds encoder change in mean neighbour replacement. Replacing mean with CLS or SEP for the four word-level BERTs reverses that latter ordering. Four representations move together in these scenarios, including the cross-model reference. The result is a sensitivity of the practical comparison, without an isolated causal effect of pooling. It gives no justification for choosing a recipe simply because it produces more agreement.

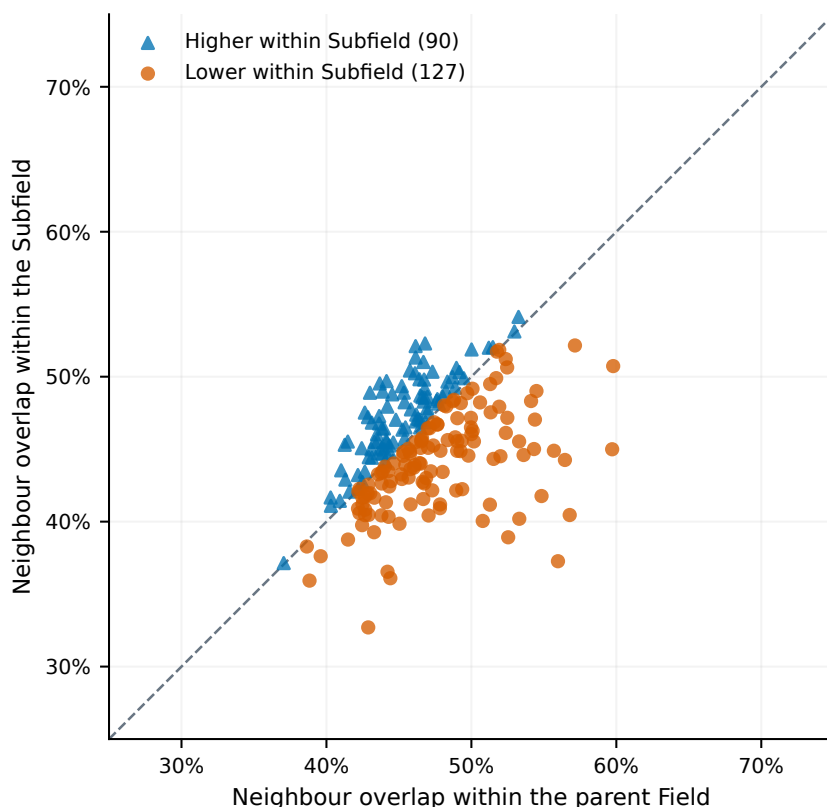


Figure 2: Neighbour agreement in matched Field and Subfield searches. Each point is one of 217 Subfields, averaged over model pairs and ten candidate selections. Both scopes contain the same 50 fixed queries within 256 candidates and matched dates; self is excluded. The broader Field pool is conditioned on these queries. Points above the diagonal have higher overlap in the narrower search; those below have lower overlap. Dispersion describes group differences, not confidence intervals.

The larger input panel reduces sensitivity to article selection. Using 100 comparable selections at each size, the shape-contrast screen drops from 72 alerts among 520 model–Field–input contrasts at 26,000 articles to five at 52,000. The earlier 20-selection check gives 31 and four, respectively (Figure S1). Both records are retained. Passing this screen does not mean that a contrast differs from zero, nor does it certify all neighbour results or readout variants. The common-fragment and MiniLM-length checks further distinguish unequal text exposure from a coding error: the original MiniLM limit was its documented setting, not an implementation fault.

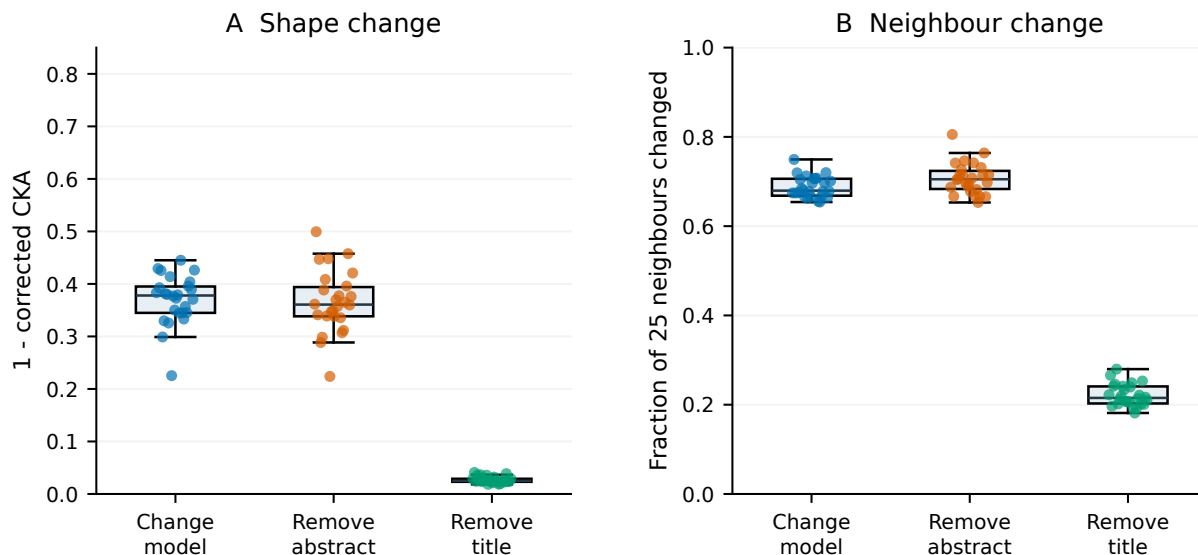


Figure 3: Changes associated with model and text input. The same 52,000 IDs provide 2,000 candidates per Field. The reference input is title plus abstract: “remove abstract” leaves the title, and “remove title” leaves the abstract. (A) One minus corrected CKA. (B) Fraction of the 25 neighbours replaced. Each point is a Field mean over the fixed model panel. Boxes show medians and interquartile ranges, with 1.5-IQR whiskers; all Field points are displayed. These are not confidence intervals. Native text limits are retained.

4.4 Persistence and reversal of geometric field comparisons

The geometric extension turns to a claim a reader might actually make: that one Field has a greater measured property than another. I first check the change in articles within a model. Across the 260 model–Field cases, angular spread has no alerts under either half-sample or equal-size selection. Effective linear dimension retains four half-sample alerts, although the five equal-size selections produce none (Table S12). Both observations belong in the account. The successful equal-size check cannot stand in for the four cases flagged by the other screen.

Changing the model exposes the distinction. Across all 325 Field pairs, angular spread has a persistent direction common to all ten models in 42 pairs, persistent opposition in 262 and an unresolved result in 21. For effective linear dimension, the counts are 54, 221 and 50 (Figure 4). In an opposition, at least two particular models maintain opposite directions over all 26 saved selection conditions. This is a claim about those persistent opposing models, not a requirement that every model disagree. An unresolved pair may still contain partial agreement.

The size of the difference changes what can be made of those totals. Many angular oppositions are small. Requiring both opposing directions to exceed a symmetric relative difference

of 5% leaves 96 angular pairs; at 10%, 19 remain. Effective linear dimension retains 177 and 126 oppositions at the same cutoffs. I keep these magnitudes beside the direction-only counts because persistent differences need not be large differences. The minimum is an explicit sensitivity choice, not a universal boundary of scientific importance. Figure 4 keeps all 325 pairs in every row, including the growing unresolved group.

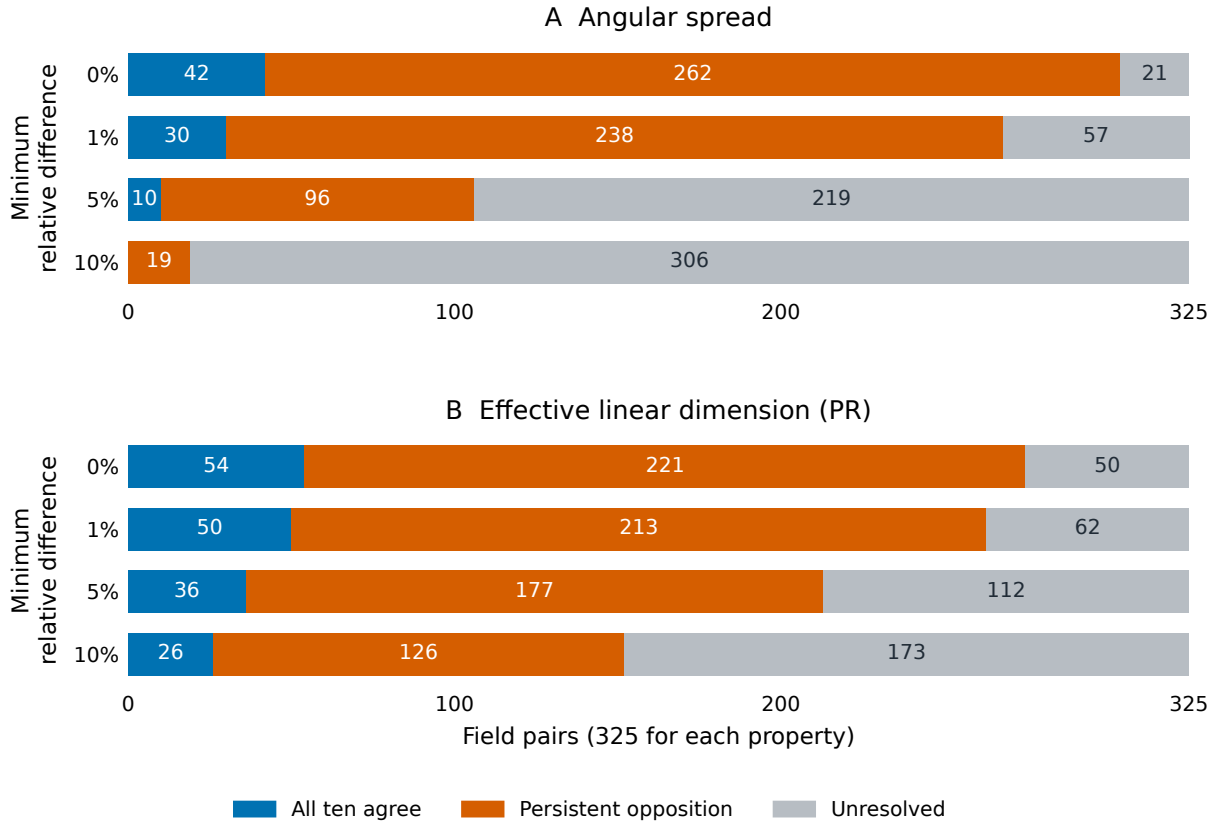


Figure 4: Persistence, opposition and magnitude of Field comparisons. Each row contains all 325 Field pairs for one property and ten models. A model direction must persist across the primary selection, twenty half samples and five additional equal-size selections from the same corpus. All-ten agreement requires a common persistent direction; opposition requires at least two fixed models with persistent opposite directions. All remaining pairs are unresolved. Magnitude uses $2(B - A)/(A + B)$. The four minima are operational cutoffs, not significance or accuracy thresholds. The 26 conditions overlap; point controls are reported separately.

Alternative measures preserve much of the pattern. Requiring the same opposing models to keep their directions across the angular alternatives retains 225 of the original 262 oppositions. The spectral alternatives retain 196 of 221. Their coverage differs: angular alternatives have all 26 selection conditions, whereas entropy-based rank and the 80%-variance count are available for three. These alternatives also share much of their mathematical construction. Their agreement supports a geometric description without turning it into a measure of topic diversity.

The result is not confined to the four word-level BERTs. Restricting the panel to SPECTER, SPECTER2, SciNCL, MPNet, MiniLM and SimCSE still yields 231 angular and 160 effective-dimension oppositions at zero minimum. A smaller panel has fewer opportunities to contain opposing models, so the change in counts does not isolate an effect of training objective. All model omissions and the complete pair classifications accompany Tables S14–S15.

Processing choices set a further boundary. After subtracting the global centre and renormalising, the same original opposing model pair and directions survive for 145 of 262 angular oppositions and 218 of 221 effective-dimension oppositions in the matched point check. Angular ordering is therefore more dependent on that choice. Pooling changes can also reverse comparisons (Table S16). These are point sensitivities with matched references, not another set of 26-repeat confirmations. The retained conclusions concern explicitly defined representations, rather than properties expected to survive any transformation.

Connection is where the proposed substantive interpretation failed the checks. Elongated clouds and isolated extreme points could resemble fragmentation under some rules, and the measurements changed substantially with sample size and neighbour count. The equal-size selection screen alone retained 51 connection alerts among 260 cases. I keep these results in Supplement S6 with the failed construct checks. They give no defensible basis here for saying that a Field contains more separate thematic groups.

5 Discussion

5.1 From map agreement to the persistence of a claim

The finding I would carry into a science-mapping application is that a repeatable comparison can still depend on its representation. The centre analysis establishes broad shared structure above the random-group reference. The Field comparisons then show persistent opposite orderings under different models, despite repeated changes of articles. A global agreement score leaves that disagreement unresolved. I would therefore attach the robustness claim to the relationship, neighbour set or Field comparison being used, and follow that claim through the alternative representations.

This is a continuation of the methodological question already raised in science mapping (Gläser et al., 2017; Boyack and Klavans, 2020). Evaluating a representation against an external classification, as Constantino et al. (2025) do, establishes performance relative to that criterion. Comparing alignment and retrieval, as in Caspari et al. (2024), establishes which relationships representations share. The contribution I claim here connects those levels with the persistence and magnitude of specific Field comparisons on matched articles. It draws on existing agreement measures and an established concern with representation choice.

There is a reason to be exact about what the geometric quantities mean. An angle describes separation, and PR describes how variance occupies directions. Interdisciplinarity and semantic diversity would add claims about the contents of papers. Those claims need their own evidence, as related work on embedding isotropy also cautions (Rudman et al., 2022). The fragmentation checks show why I keep this boundary explicit. A measurement can repeatedly distinguish two clouds while failing to establish the substantive difference that its name appears to promise.

5.2 Implications for comparative science mapping

Before interpreting a science map, I would write down the comparison it is expected to support. For retrieval, that comparison includes the candidate papers and the number of neighbours. For Field geometry, it includes the representation rule, article balance and minimum difference. Those choices give the robustness check a concrete object. Reporting should be equally specific: a model name needs its title/abstract input, text limit and pooling rule. Otherwise, a reader knows which encoder was used but cannot reconstruct the representation from that information alone.

Medicine is a useful reason to resist a single explanation of disagreement. Its shape agreement is below Energy’s, yet its neighbour agreement is not consistently the lowest. Balancing Subfield composition and excluding BioBERT and PubMedBERT both leave the Medicine–Energy difference (Supplement S5). Those results constrain explanations without selecting a unique cause. Model families also shift with readout. Since these encoders differ in linked architectures, training sources and objectives, their similarities cannot disentangle those influences causally.

The temporal comparison illustrates how a conclusion can turn on the comparison opportunity. Mean shape agreement rises slightly from the first period to the last. Neighbour agreement falls by 0.87 percentage points with all candidates and rises by 0.91 points with 2,048 candidates per group. The search design changes the sign. I do not read this as historical convergence of science. Current models represent publications from different dates using their current parameters; they do not recover the knowledge available at the time those publications appeared.

5.3 Limits and next validation

A large corpus still has boundaries. This one contains selected OpenAlex records with usable abstracts and predominantly English text. The reinforcement sample, balanced panels and equal weights support comparisons across Fields without estimating the worldwide distribution of research. The atlas leaves visible classification and content errors that survived the cleaning rules. Removing specified problems cannot certify every record. Saved selections

and frozen checkpoints let others inspect the comparison; further selections from the same corpus leave its missing coverage untouched.

The model panel needs the same qualification. The encoders share some architectures, training sources and objectives, and several were not trained specifically for similarity. Checks restricted to six similarity-oriented encoders and alternative readouts help locate sensitivities within the panel, without turning it into a representative sample of all encoders. A persistent opposition establishes that the choice matters among these models. It cannot decide which of the opposing orderings is substantively right.

I also retain the exploratory history. The geometry extension followed earlier results, and the all-pairs summary followed its initial examples. Making that sequence and the complete tables available does not turn the later comparisons into a confirmatory study. The next question I would pursue is whether one consequential field comparison agrees with independently assessed relationships between articles or answers a discipline-specific question. That step would test the interpretation left open here. A larger model panel alone would not supply it.

6 Conclusion

A shared broad map leaves room for different neighbours and opposite geometric comparisons between Fields. Text input can matter as much as model choice, and a result that survives changing the articles may still reverse when the encoder changes. I would therefore assess robustness where the scientific claim is made. Angular spread and effective linear dimension allow that assessment for defined geometric properties, with magnitudes and processing sensitivities kept visible. The distinction that remains is essential: these are properties of the representations. Establishing what they say about scientific knowledge requires further validation.

Declarations

Author contributions. Alejandro Treny Ortega conceived the project and directed the research and manuscript preparation as an independent researcher. Computational and editorial assistance is disclosed below.

Competing interests. The author declares no competing interests.

Funding. This research received no external funding.

Data and code availability. The project repository is <https://github.com/aleetreny/Shape-of-Science-Robustness>. This draft is accompanied by the tables used for

its figures and the editable manuscript sources. The frozen corpus, vectors and complete local execution records have not been deposited in a persistent public archive. Public code alone does not provide full reproduction of the present analysis. The access inventory and remaining release requirements are stated in Supplement S8; no data deposit or persistent identifier is claimed here.

Use of AI-assisted tools. OpenAI Codex assisted with literature searches, software development and analysis, visualisation, and manuscript drafting and editing. The author is responsible for the content.

References

- Iz Beltagy, Kyle Lo, and Arman Cohan. SciBERT: A pretrained language model for scientific text. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3615–3620, Hong Kong, China, nov 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1371. URL <https://aclanthology.org/D19-1371/>.
- Kevin W. Boyack and Richard Klavans. A comparison of large-scale science models based on textual, direct citation and hybrid relatedness. *Quantitative Science Studies*, 1(4):1570–1585, 2020. ISSN 2641-3337. doi: 10.1162/qss_a_00085.
- Laura Caspari, Kanishka Ghosh Dastidar, Saber Zerhoubi, Jelena Mitrovic, and Michael Granitzer. Beyond benchmarks: Evaluating embedding model similarity for retrieval augmented generation systems. arXiv:2407.08275, 2024. URL <https://arxiv.org/abs/2407.08275>. Preprint.
- Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2270–2282, Online, jul 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.207. URL <https://aclanthology.org/2020.acl-main.207/>.
- Isabel Constantino, Sadamori Kojaku, Santo Fortunato, and Yong-Yeol Ahn. Representing the disciplinary structure of physics: A comparative evaluation of graph and text embedding methods. *Quantitative Science Studies*, 6:263–280, 2025. ISSN 2641-3337. doi: 10.1162/qss_a_00349.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Tamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, jun 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL <https://aclanthology.org/N19-1423/>.
- Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-

- tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910, Online and Punta Cana, Dominican Republic, nov 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.552. URL <https://aclanthology.org/2021.emnlp-main.552/>.
- Jochen Gläser, Wolfgang Glänzel, and Andrea Scharnhorst. Same data—different results? towards a comparative approach to the identification of thematic structures in science. *Scientometrics*, 111(2):981–998, 2017. ISSN 0138-9130. doi: 10.1007/s11192-017-2296-z.
- Rita González-Márquez, Luca Schmidt, Benjamin M. Schmidt, Philipp Berens, and Dmitry Kobak. The landscape of biomedical research. *Patterns*, 5(6):100968, 2024. ISSN 2666-3899. doi: 10.1016/j.patter.2024.100968.
- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2022. ISSN 2691-1957. doi: 10.1145/3458754.
- Matthias Held and Theresa Velden. How to interpret algorithmically constructed topical structures of scientific fields? a case study of citation-based mappings of the research specialty of invasion biology. *Quantitative Science Studies*, 3(3):651–671, 2022. ISSN 2641-3337. doi: 10.1162/qss_a_00194.
- Nathaniel Imel and Zachary Hafen. Density, asymmetry and citation dynamics in scientific literature. arXiv:2506.23366, 2025. URL <https://arxiv.org/abs/2506.23366>. Preprint.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/kornblith19a.html>.
- Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020. ISSN 1367-4803. doi: 10.1093/bioinformatics/btz682.
- Alex Murphy, Joel Zylberberg, and Alona Fyshe. Correcting biased centered kernel alignment measures in biological and artificial neural networks. arXiv:2405.01012, 2024. URL <https://arxiv.org/abs/2405.01012>. Preprint.
- OpenAlex. Topics. OpenAlex Help Center, 2026. URL <https://help.openalex.org/data/topics/>. Accessed 18 September 2026.
- Malte Ostendorff, Nils Rethmeier, Isabelle Augenstein, Bela Gipp, and Georg Rehm. Neighborhood contrastive learning for scientific document representations with citation embeddings. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11670–11688, Abu Dhabi, United Arab Emirates, dec 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.802. URL <https://aclanthology.org/2022.emnlp-main.802/>.

- Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts, 2022. URL <https://arxiv.org/abs/2205.01833v2>. Preprint, version 2.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China, nov 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1410. URL <https://aclanthology.org/D19-1410/>.
- William Rudman, Nate Gillman, Taylor Rayne, and Carsten Eickhoff. IsoScore: Measuring the uniformity of embedding space utilization. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3325–3339, Dublin, Ireland, may 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.262. URL <https://aclanthology.org/2022.findings-acl.262/>.
- Amanpreet Singh, Mike D’Arcy, Arman Cohan, Doug Downey, and Sergey Feldman. SciRepEval: A multi-format benchmark for scientific document representations. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5548–5566, Singapore, dec 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.338. URL <https://aclanthology.org/2023.emnlp-main.338/>.
- Shruti Singh and Mayank Singh. The inefficiency of language models in scholarly retrieval: An experimental walk-through. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3153–3173, Dublin, Ireland, may 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.249. URL <https://aclanthology.org/2022.findings-acl.249/>.
- Alex H Williams. Equivalence between representational similarity analysis, centered kernel alignment, and canonical correlations analysis. In Marco Fumero, Clementine Domine, Zorah Löhner, Donato Crisostomi, Luca Moschella, and Kimberly Stachenfeld, editors, *Proceedings of UniReps: the Second Edition of the Workshop on Unifying Representations in Neural Models*, volume 285 of *Proceedings of Machine Learning Research*, pages 10–23. PMLR, 14 Dec 2024. URL <https://proceedings.mlr.press/v285/williams24a.html>.
- Alex H Williams, Erin Kunz, Simon Kornblith, and Scott Linderman. Generalized shape metrics on neural representations. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 4738–4750. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/252a3dbaeb32e7690242ad3b556e626b-Paper.pdf.